

Root Cause Analysis Report

Prepared for: Bajaj Earths

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Subject: Recurring Furnace Backfire & Visible Stack Emissions

Process: Calcium Bentonite Ore Thermal Activation (Coal-Fired Rotary Kiln, 10 TPH)

REFERENCE DOCUMENT

All equipment specifications, gas volume calculations, and duct sizing data referenced in this report are detailed in the companion: **LIRN - System Reference - Bajaj Earths Kiln Bag Filter**.

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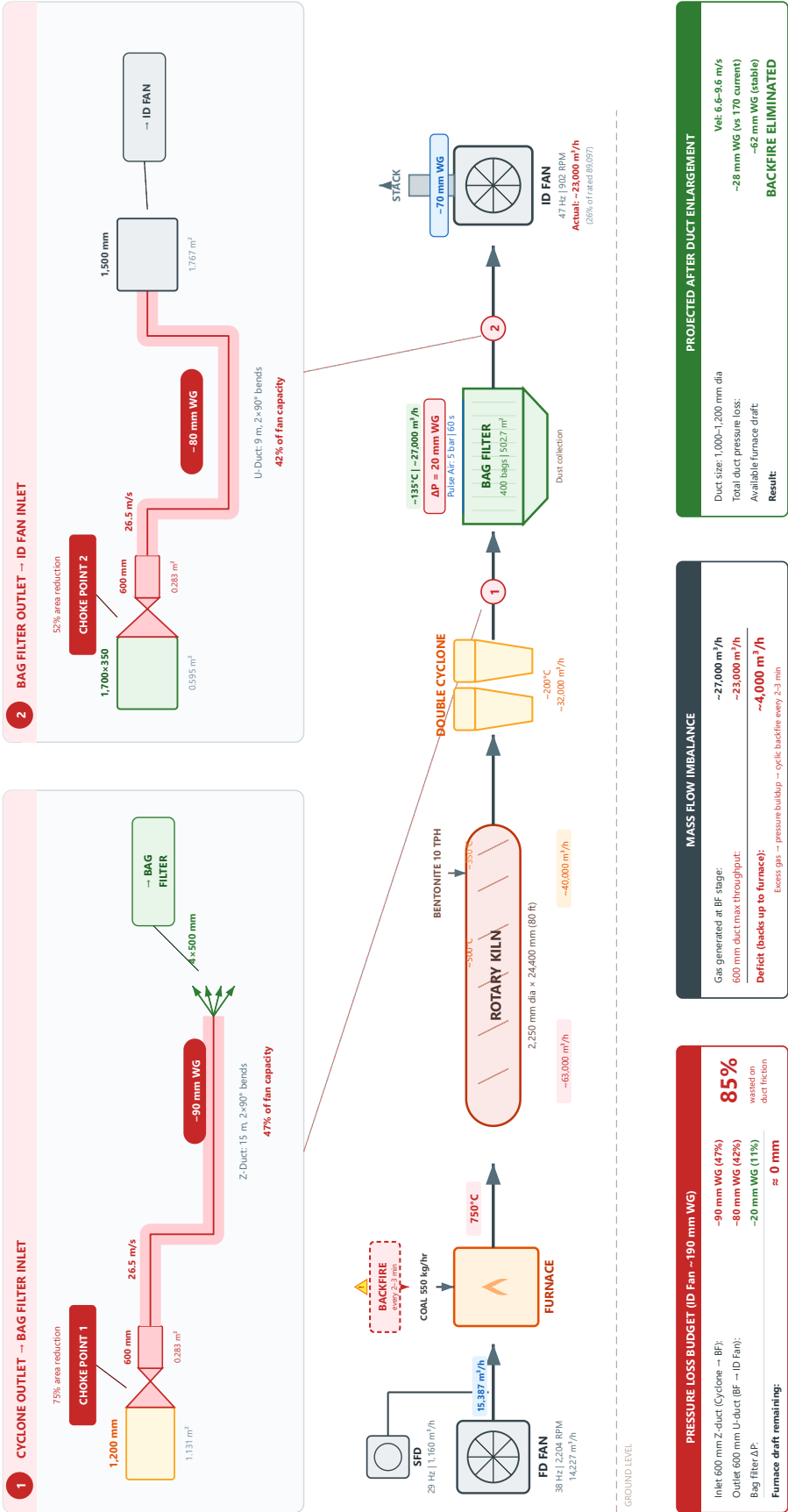
1. Executive Summary

The calcium bentonite ore processing plant is experiencing two operational failures: **cyclic furnace backfire every 2-3 minutes** and **visible fine dust emissions from the exhaust stack**.

Our investigation identifies a single structural root cause driving both issues: **two 600 mm diameter duct sections** (connecting the cyclone to the bag filter inlet, and the bag filter to the ID fan outlet) are severely undersized for the gas volume generated by the process. These restrictions consume **85% of the ID fan's total draft capacity as friction loss**, leaving near-zero negative pressure at the furnace. This causes cyclic positive-pressure blowback (backfire). The restricted flow also contributes to inadequate dust cake formation in the bag filter, allowing fine particles to pass through to the stack.

Primary corrective action: Enlarge both 600 mm duct sections to 1,000-1,200 mm diameter. This structural modification recovers **~142 mm WG of draft pressure**, eliminates backfire, enables proper bag filter operation, and unlocks ~74% energy savings on the ID fan.

2. System Layout



* Diagram is representational and not to scale. Circled numbers reference duct detail views above.

For full specifications, see the companion System Reference Document.

3. Problem Statement

3.1 Observed Symptoms

#	Symptom	Details
1	Furnace Backfire	Hot gas erupts from furnace inlet every 2-3 minutes; subsides for ~1 minute; cycle repeats continuously
2	Visible Stack Emissions	Fine calcium bentonite dust visible in exhaust; product loss and environmental concern
3	Low Bag Filter DP	Differential pressure across bags measured at 20 mm WG (expected: 80-130 mm WG)
4	High Outlet Vacuum	-70 mm WG measured at bag filter outlet vs. atmosphere

3.2 Troubleshooting Already Performed

Action Taken	Result	Conclusion
Reduced ID fan VFD frequency	Backfire frequency increased	Confirms draft deficiency, not excess flow
Stopped pulse cleaning for 8 hours	No change in backfire pattern	Rules out pulse-induced pressure waves as cause
Inspected all bag collars and bags	All intact, no leaks found	Rules out bag or seal failure
Reduced FD fan	Not viable; combustion requires 750°C kiln inlet temperature	FD airflow cannot be reduced

4. Root Cause Analysis

4.1 PRIMARY — Furnace Backfire: Undersized Ductwork Strangling System Draft

The Problem

The gas path between the cyclone and the ID fan passes through **two 600 mm diameter duct sections** that are severely undersized relative to the upstream and downstream connections they serve.

Duct Section	Route	Upstream Connection	Restriction	Area Reduction
Cyclone → Bag Filter	Z-route: 15 m, 2x90° bends, 5 m drop	1,200 mm dia (1.131 m ²)	600 mm dia (0.283 m²)	75%
Bag Filter → ID Fan	U-route: 9 m, 2x90° bends, 2.5 m drop	1,700x350 mm rect (0.595 m ²)	600 mm dia (0.283 m²)	52%

How This Causes Backfire

The ID fan's role is to create **negative pressure (vacuum)** throughout the system, pulling hot gas from the furnace forward through the kiln, cyclone, bag filter, and out the stack. For stable furnace operation, the furnace must maintain negative draft (typically -10 to -30 mm WG) so that combustion gas always flows forward.

The ID fan (rated 89,097 m³/h at 902 RPM) has more than enough capacity for this. However, **the 600 mm ducts act as a bottleneck that consumes almost all the fan's pressure capability before it can reach the furnace:**

Pressure Loss Budget, Current System:

Component	Pressure Loss (mm WG)	Share of ID Fan Capacity
600 mm inlet duct (Z-route: 15 m, 2x90° bends, 5 m drop)	~90	47%
600 mm outlet duct (U-route: 9 m, 2x90° bends, 2.5 m drop)	~80	42%
Bag filter (across bags)	20	11%
Total system loss	~190	100%
ID fan total pressure capacity	~190	—
Remaining for furnace draft	~0	0%

With zero draft remaining at the furnace, the system operates at the threshold of positive pressure. Meanwhile, gas generation is continuous:

- Coal combustion produces ~3,300 m³/h of flue gas
- Moisture evaporation adds ~4,050 m³/h of steam (at 135°C)
- Combined with input air, **~27,000 m³/h** of gas arrives at the bag filter stage

But the 600 mm ducts can only pass **~23,000 m³/h**, leaving a **4,000 m³/h deficit**. The excess gas accumulates, pressure builds backward through the cyclone and kiln to the furnace, the furnace goes positive pressure, and **hot gas erupts backward as backfire**. This relieves the overpressure, vacuum momentarily recovers, and the cycle restarts every 2-3 minutes.

Why Reducing ID Fan Speed Made It Worse

The 600 mm ducts impose a hard ceiling on gas throughput regardless of fan speed. Reducing the VFD frequency below 47 Hz lowers the fan's pressure capability, meaning it has *even less* vacuum to overcome the duct friction. Less gas gets through, pressure builds faster, and backfire becomes more frequent. **The bottleneck is the duct, not the fan.**

At its current 47 Hz, the ID fan delivers only **~23,000 m³/h actual flow (just 26% of its rated 89,097 m³/h capacity)** because it is operating against an excessively high system resistance curve imposed by the undersized ducts.

4.2 SECONDARY — Stack Emissions: Premature Dust Cake Removal

In bag filtration, the bag fabric itself is a coarse filter. The actual fine-particle filtration is performed by the **dust cake**, the accumulated layer of material on the bag surface. A mature dust cake typically produces a differential pressure of 80-130 mm WG.

The current system uses a **60-second pulse cleaning timer**, which strips the dust cake before it can mature:

Indicator	Measured	Expected (Healthy)	Interpretation
Bag filter DP	20 mm WG	80-130 mm WG	No mature dust cake present
Bag integrity	All intact	N/A	Fabric is fine; issue is bare fabric cannot catch fines
Stack emissions	Visible dust	Clear	Fine bentonite passes through uncaked bags

The duct restriction is also a contributing factor: the restricted flow creates uneven gas distribution across the bag house, further degrading filtration efficiency.

5. Before & After — Corrective Action Impact

5.1 Duct Modification Impact (P1 + P2)

Parameter	CURRENT (600 mm)	AFTER FIX (1,000-1,200 mm)	Change
Inlet duct diameter	600 mm	1,000-1,200 mm	+67-100%
Inlet duct cross-section	0.283 m ²	0.785-1.131 m ²	+177-300%
Inlet gas velocity	26.5 m/s	6.6-9.6 m/s	-64-75%
Inlet duct pressure loss	~90 mm WG	~8-13 mm WG	-86-91%
Outlet duct diameter	600 mm	800-1,000 mm	+33-67%
Outlet duct cross-section	0.283 m ²	0.503-0.785 m ²	+78-177%
Outlet gas velocity	26.5 m/s	9.6-14.9 m/s	-44-64%
Outlet duct pressure loss	~80 mm WG	~10-15 mm WG	-81-88%
Total duct friction	~170 mm WG	~18-28 mm WG	-84-89%
Available furnace draft	~0 mm WG	~62-72 mm WG	Stable operation
Backfire	Every 2-3 min	Eliminated	—

5.2 Pulse Cleaning Adjustment Impact (P3)

Parameter	CURRENT	AFTER FIX
Trigger mode	Timer (60 sec)	DP-triggered at 120-130 mm WG
Fallback timer	None	300-600 sec (if no DP sensor)
Pulse pressure	5 bar	Test at 3-3.5 bar
Bag filter DP	20 mm WG	80-130 mm WG (healthy cake)
Stack emissions	Visible dust	Below visible threshold

5.3 Overall System Comparison

Parameter	Current	Proposed
Inlet duct loss	90 mm WG (47%)	13 mm WG (7%)
Bag filter DP	20 mm WG (11%)	100 mm WG (53%)
Outlet duct loss	80 mm WG (42%)	15 mm WG (8%)
Total system loss	190 mm WG	128 mm WG
Furnace draft	~0 mm WG (BACKFIRE)	~62 mm WG (STABLE)
ID fan VFD	47 Hz	25-32 Hz
ID fan power	100%	~26% (74% saving)
Gas throughput	~23,000 m ³ /h (choked)	~27,000 m ³ /h (full flow)
Stack emissions	VISIBLE DUST	CLEAN

6. Recommendations

Priority	Action	Addresses	Details
P1	Enlarge inlet duct (Cyclone → Bag Filter)	Backfire	Replace 600 mm with 1,000-1,200 mm dia for full 15 m Z-route. The cyclone outlet is already 1,200 mm; this undoes an unnecessary restriction. Recovers ~77 mm WG draft.
P2	Enlarge outlet duct (Bag Filter → ID Fan)	Draft balance	Replace 600 mm with 800-1,000 mm dia for full 9 m U-route. Recovers ~65 mm WG draft.
P3	Switch pulse cleaning to DP-triggered	Emissions	Set DP trigger at 120-130 mm WG threshold. If DP sensor unavailable, extend timer to 300-600 sec as interim. Test reducing pulse pressure from 5 bar to 3-3.5 bar.
P4	Reduce ID fan VFD frequency	Energy savings	After P1+P2 commissioning, reduce from 47 Hz to 25-32 Hz. Power savings: $(32/47)^3 = 0.31$, approximately 69-74% energy reduction.

Implementation sequence: P1 and P2 should be executed together during a single planned shutdown. P3 can be configured during P1+P2 commissioning. P4 is tuned incrementally during post-modification start-up.

Appendix A: Calculated Gas Volume at Each Process Stage

A.1 Gas Generation Sources

Source	Calculation	Volume (m ³ /h)
Combustion air input (FD + SFD)	14,227 + 1,160	15,387
Coal combustion flue gas	550 kg/hr × ~6 m ³ /kg (products at temperature)	~3,300
Moisture evaporation (steam)	3,158 kg/hr water evaporated → steam at 135°C	~4,050

Coal parameters: 550 kg/hr average feed, GCV 4,200 kcal/kg, heat input 2,310,000 kcal/hr.
Theoretical air requirement: ~5.5 m³/kg coal. Actual air supply: 15,387 m³/h. Excess air ratio: ~5:1.
Moisture evaporated: Feed rate 10 TPH × (35% input – 5% output) = 3,158 kg/hr water removed.

A.2 Gas Volume at Each Stage

Gas volume increases with temperature per ideal gas law: $V_{\text{hot}} = V_{\text{cold}} \times (T_{\text{hot}} + 273) / (T_{\text{cold}} + 273)$

Stage	Temp (°C)	Gas Volume (m ³ /h)	Notes
FD + SFD air input	~30	15,387	Ambient air, measured at fans
Furnace exit / Kiln inlet	~750	~63,000-68,000	Combustion gas + heated excess air + early steam
Kiln outlet	~350	~38,000-42,000	After heat transfer to bentonite material
Cyclone inlet	~200	~30,000-34,000	Further cooling through ductwork
Cyclone outlet / Bag filter inlet	~135	~25,000-28,000	Design flow for the bag filter circuit

A.3 Duct Velocity at Each Cross-Section

Calculated at ~27,000 m³/h (mid-range gas volume at bag filter stage):

Location	Size	Area (m ²)	Velocity (m/s)	Limit	Status
Cyclone outlet	1,200 mm dia	1.131	6.6	18 m/s	OK
Inlet duct	600 mm dia	0.283	26.5	18 m/s	CHOKED
Bag filter inlets	4 × 500 mm dia	0.785	9.6	18 m/s	OK
Bag filtration (A/C)	502.7 m ² total	—	0.90 m/min	1.5 m/min	OK
BF rectangular outlet	1,700 × 350 mm	0.595	12.6	18 m/s	OK
Outlet duct	600 mm dia	0.283	26.5	18 m/s	CHOKED
ID fan inlet	1,500 mm dia	1.767	4.2	18 m/s	OK

Recommended maximum duct velocity for mineral dust-laden gas: 15-18 m/s. Both 600 mm sections exceed this by 47-77%.

Appendix B: Pressure Loss Calculation Detail

B.1 Inlet Duct — 600 mm Z-Route (Cyclone → Bag Filter)

Route: 1,200 mm cyclone outlet → contraction to 600 mm → 15 m total run with 2 × 90° bends → 5 m altitude drop → expansion to 4 × 500 mm bag filter inlets.

Component	Basis	Loss (mm WG)
Contraction 1,200 → 600 mm	$K \approx 0.3$ at 26.5 m/s	15-20
Straight run: 15 m at 26.5 m/s	$f \approx 0.02$, $D = 0.6$ m	25-35
90° bend × 2	$K \approx 0.5$ each at 26.5 m/s	36-44
Expansion to 4 × 500 mm	$K \approx 0.15$	5-8
Subtotal		81-107
Value used in analysis		~90

B.2 Outlet Duct — 600 mm U-Route (Bag Filter → ID Fan)

Route: 1,700 × 350 mm rectangular BF outlet → contraction to 600 mm → 9 m total run with 2 × 90° bends → 2.5 m altitude drop → expansion to 1,500 mm ID fan inlet.

Component	Basis	Loss (mm WG)
Contraction rect → 600 mm	$K \approx 0.25$ at 26.5 m/s	12-18
Straight run: 9 m at 26.5 m/s	$f \approx 0.02$, $D = 0.6$ m	15-22
90° bend × 2	$K \approx 0.5$ each at 26.5 m/s	36-44
Expansion to 1,500 mm	$K \approx 0.15$	5-8
Subtotal		68-92
Value used in analysis		~80

B.3 ID Fan Operating Point

Parameter	Value
Rated airflow	89,097 m ³ /h
Rated RPM	960
Current VFD frequency	47 Hz
Operating RPM	$960 \times (47/50) = \mathbf{902 \text{ RPM}}$
Total pressure at 902 RPM	~190 mm WG
Actual system throughput	~23,000 m ³ /h
Fan capacity utilization	26% of rated

----- End of Report -----

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